Majbah Uddin

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Leads multidisciplinary teams in multiyear, externally sponsored transportation research that integrates diverse data and decision science.

Research Areas and Interests

AI and Machine Learning	Multi-agent AI for transportation operations, machine learning for transportation and logistics, decision support
Resilience	Transportation system resilience to extreme weather events, supply chain disruption and recovery
Freight and Supply Chains	Freight flow modeling, national freight data (FAF), intermodal networks, truck size and weight
Decision Science	Optimization and operations research under uncertainty, agent-based and game-theoretic modeling
Mobility and Data	Travel behavior, transportation planning, data fusion and analytics

Professional Experience

Oak Ridge National Laboratory — *Oak Ridge, TN*

R&D Staff — *March 2024–Present*

R&D Associate Staff — *December 2021–March 2024*

- Lead multidisciplinary teams as PI or co-PI on externally sponsored and internal research projects totaling more than $15M, for DOE, USDOT (FHWA, BTS), NYSDOT, and ORNL LDRD
- Manage federal projects, including interagency agreements and Strategic Partnership Projects, through scoping, budgets, milestones, and deliverables
- Develop significant new research proposals and sponsor relationships, leading to new multiyear, multi-institution projects
- Lead development of national freight data and models, including the Freight Analysis Framework (FAF) releases used by federal and state DOTs, national labs, and industry
- Translate research into decision-support tools and software for federal and state agencies
- Integrate expertise across transportation, energy, water, and earth-system sciences, coordinating staff across ORNL directorates to solve complex, cross-disciplinary problems
- Serve as a primary technical contact for federal and state sponsors and help shape research directions in AI and transportation resilience through selected project ideas, invited workshops, and panels
- Mentor and direct the research of postdoctoral researchers, technical staff, and student interns

Oak Ridge National Laboratory — *Oak Ridge, TN*

Postdoctoral Research Associate — *September 2019–November 2021*

- Produced freight flow datasets for several out-of-scope components of the FAF version 5, not covered by the Commodity Flow Survey (CFS)
- Generated regional travel statistics from National Household Travel Survey (NHTS) data for NYSDOT regional travel demand modeling
- Compiled a national pipeline intermodal terminal dataset by merging multiple sources

University of South Carolina — *Columbia, SC*

Postdoctoral Research Associate — *May 2019–September 2019*

University of South Carolina — *Columbia, SC*

Graduate Research Assistant — *August 2013–April 2019*

University of South Carolina — *Columbia, SC*

Graduate Teaching Assistant — *August 2017–April 2019*

Stamford University Bangladesh — *Dhaka, Bangladesh*

Lecturer — *February 2013–August 2013*

Research Project Leadership

2026–2031	**PI,** *Research for Truck Size and Weight Limits*, Federal Highway Administration/USDOT, $6.5M
2026–2029	**PI,** *Multi-Agent AI System for Resilient and Energy-Efficient Regional Transportation Operations*, DOE Transportation Technologies Office, $2.5M
2026–2028	**PI,** *Coupled Water, Freight, and Passenger Resilience Modeling for the Southeast*, ORNL Laboratory Directed Research and Development, $1.0M
2026–2029	**PI,** *Corridor-Level National Freight Flow Model Development*, Federal Highway Administration/USDOT, $1.0M
2024–2026	**Co-PI,** *Mining Waste–Driven Carbon Dioxide Capture and Storage*, ORNL Laboratory Directed Research and Development, $875K
2023–2028	**Co-PI (2023–2025); PI (2026–2028),** *Freight Analysis Framework (FAF) Data and Model Development Support*, Bureau of Transportation Statistics/USDOT, $4.2M
2023–2025	**Co-PI,** *Coupled Ecosystem-and-Engineering Decision-Making Framework for Enhanced Weathering*, ORNL Laboratory Directed Research and Development, $1.2M
2023–2025	**PI,** *Non-Highway Gasoline and Special Fuel Consumption Estimation Model Study*, Federal Highway Administration/USDOT, $320K
2022–2023	**Co-PI (2022); PI (2023),** *Modeling a Gigaton CO_2 Economy*, ORNL Laboratory Directed Research and Development, $490K
2020–2025	**Team Member (2020–2021); PI (2022–2025),** *Research, Analysis, and Program Support for NYSDOT*, New York State Department of Transportation, $3.8M

Education

University of South Carolina — *Columbia, SC*

PhD in Civil Engineering — *January 2016–May 2019*

- Dissertation: *Development of Models for Road-Rail Intermodal Freight Network under Uncertainty*
- Major: Transportation Engineering and Operations Research

University of South Carolina — *Columbia, SC*

Master of Applied Statistics — *August 2016–May 2018*

University of South Carolina — *Columbia, SC*

MS in Civil Engineering — *August 2013–December 2015*

- Thesis: *Assessment of Classifiers for Potential Voice-Enabled Transportation Apps*
- Major: Transportation Engineering

Bangladesh University of Engineering and Technology — *Dhaka, Bangladesh*

BS in Civil Engineering — *June 2007–April 2012*

Journal Articles

Google Scholar Metrics Citations: 1216 h-index: 16 i10-index: 19 Underlined (throughout): postdoc, student, or staff mentee

1. Jiafu Mao, Yuefeng Hao, David Sholl, Xin Gu, Yaoping Wang, Andrew G Stack, **Majbah Uddin**, Xiaoying Shi, Nagendra Singh, Dan Lu, Ingrid Busch, Fengming Yuan, Mingzhou Jin, David McCollum, Daniel M Ricciuto, and Peter E Thornton. Forthcoming. "Enhanced Rock Weathering for Sustainable Carbon Capture: Advances, Challenges, and Future Prospects in Terrestrial Ecosystems." *Reviews in Environmental Science and Bio/Technology*.
2. Nicholas Gregorich, Joseph Kalla, Mary Hannah Irwin, Saurabh Prakash Pethe, Mustafa Abbas, Zachary Coin, Mariappan Parans Paranthaman, **Majbah Uddin**, Shailesh Dangwal, Ramesh Bhave, and Syed Z Islam. 2026. "Mining Waste-Driven Carbon Capture via Ocean Alkalinity Enhancement." *Carbon Capture Science & Technology* 19, 100629. doi:10.1016/j.ccst.2026.100629.
3. Melrose Pan, Latif Patwary, and **Majbah Uddin**. 2025. "Unlocking Nighttime Mobility: Land Use and Accessibility in Public Transit for Night Commuters." *Journal of Transport Geography* 127, 104290. doi:10.1016/j.jtrangeo.2025.104290.
4. **Majbah Uddin**, Robin Clark, Michael Hilliard, Josh Thompson, Matthew Langholtz, and Erin Webb. 2025. "Agent-Based Modeling for Multimodal Transportation of CO_2 for Carbon Capture, Utilization, and Storage: CCUS-Agent." *Applied Energy* 378 (Part A), 124833. doi:10.1016/j.apenergy.2024.124833.
5. Melrose Pan, **Majbah Uddin**, and Hyeonsup Lim. 2024. "Understanding Electric Vehicle Ownership Using Data Fusion and Spatial Modeling." *Transportation Research Part D: Transport and Environment* 127, 104075. doi:10.1016/j.trd.2024.104075.

6. **Majbah Uddin**, Nathan Huynh, and Fahim Ahmed. 2024. "Assignment of Freight Traffic in a Large-Scale Intermodal Network under Uncertainty." *Highlights of Sustainability* 3(1), 1–15. doi:10.54175/hsustain3010001.
7. Diyi Liu, Hyeonsup Lim, **Majbah Uddin**, Yuandong Liu, Lee D Han, Ho-Ling Hwang, and Shih-Miao Chin. 2024. "Improving the Accuracy of Freight Mode Choice Models: A Case Study Using the 2017 CFS PUF Data Set and Ensemble Learning Techniques." *Expert Systems with Applications* 240, 122478. doi:10.1016/j.eswa.2023.122478.
8. Yuandong Liu, **Majbah Uddin**, Shih-Miao Chin, Ho-Ling Hwang, and Jiaoli Chen. 2024. "Examining Rail Transportation Route of Crude Oil in the United States Using Crowdsourced Social Media Data." *Transportation Research Record* 2678(1), 218–228. doi:10.1177/03611981231170632.
9. Adam Siekmann, Vivek Sujan, **Majbah Uddin**, Yuandong Liu, and Fei Xie. 2024. "Optimizing Hydrogen Fueling Infrastructure Plans on Freight Corridors for Heavy-Duty Fuel Cell Electric Vehicles." *SAE International Journal of Sustainable Transportation, Energy, Environment, & Policy* 5(1), 89–111. doi:10.4271/13-05-01-0008.
10. **Majbah Uddin**, Melrose Pan, and Ho-Ling Hwang. 2023. "Factors Influencing Mode Choice of Adults with Travel-Limiting Disability." *Journal of Transport & Health* 33, 101714. doi:10.1016/j.jth.2023.101714.
11. **Majbah Uddin**, Ho-Ling Hwang, and Md Sami Hasnine. 2023. "An Interpretable Machine Learning Framework to Understand Bikeshare Demand before and during the COVID-19 Pandemic in New York City." *Transportation Planning and Technology* 46(4), 482–498. doi:10.1080/03081060.2023.2201280.
12. Hyeonsup Lim, **Majbah Uddin**, Yuandong Liu, Shih-Miao Chin, and Ho-Ling Hwang. 2022. "A Comparative Study of Machine Learning Algorithms for Industry-Specific Freight Generation Model." *Sustainability* 14(22), 15367. doi:10.3390/su142215367.
13. **Majbah Uddin**, Sabreena Anowar, and Naveen Eluru. 2021. "Modeling Freight Mode Choice Using Machine Learning Classifiers: A Comparative Study Using Commodity Flow Survey (CFS) Data." *Transportation Planning and Technology* 44(5), 543–559. doi:10.1080/03081060.2021.1927306.
14. **Majbah Uddin** and Nathan Huynh. 2020. "Injury Severity Analysis of Truck-Involved Crashes under Different Weather Conditions." *Accident Analysis & Prevention* 141, 105529. doi:10.1016/j.aap.2020.105529.
15. **Majbah Uddin** and Nathan Huynh. 2020. "Model for Collaboration among Carriers to Reduce Empty Container Truck Trips." *Information* 11(8), 377. doi:10.3390/info11080377.
16. **Majbah Uddin** and Nathan Huynh. 2019. "Reliable Routing of Road-Rail Intermodal Freight under Uncertainty." *Networks and Spatial Economics* 19(3), 929–952. doi:10.1007/s11067-018-9438-6.
17. **Majbah Uddin** and Nathan Huynh. 2018. "Factors Influencing Injury Severity of Crashes Involving HAZMAT Trucks." *International Journal of Transportation Science and Technology* 7(1), 1–9. doi:10.1016/j.ijtst.2017.06.004. **Most Cited Article Award**
18. **Majbah Uddin** and Fahim Ahmed. 2018. "Pedestrian Injury Severity Analysis in Motor Vehicle Crashes in Ohio." *Safety* 4(2), 20. doi:10.3390/safety4020020. **Best Paper Award**
19. **Majbah Uddin**, Robert Allen, Nathan Huynh, Jose Vidal, Kevin Taaffe, Lawrence Fredendall, and Joel Greenstein. 2018. "Operating Room Turnover Time Autocorrelation while using Mobile Applications." *mHealth* 4, 38. doi:10.21037/mhealth.2018.09.07.
20. **Majbah Uddin**, Robert Allen, Nathan Huynh, Jose Vidal, Kevin Taaffe, Lawrence Fredendall, and Joel Greenstein. 2018. "Assessing Operating Room Turnover Time via the Use of Mobile Application." *mHealth* 4, 12. doi:10.21037/mhealth.2018.05.03.
21. **Majbah Uddin** and Nathan Huynh. 2017. "Truck-Involved Crashes Injury Severity Analysis for Different Lighting Conditions on Rural and Urban Roadways." *Accident Analysis & Prevention* 108, 44–55. doi:10.1016/j.aap.2017.08.009.
22. Mostaqur Rahman, **Majbah Uddin**, and Sarah Gassman. 2017. "Pavement Performance Evaluation Models for South Carolina." *KSCE Journal of Civil Engineering* 21(7), 2695–2706. doi:10.1007/s12205-017-0544-7.
23. **Majbah Uddin**, Robert Allen, Nathan Huynh, Jose Vidal, Kevin Taaffe, Lawrence Fredendall, and Joel Greenstein. 2017. "Effectiveness of a Countdown Timer in Reducing OR Turnover Time." *Journal of Mobile Technology in Medicine* 6(3), 25–33. doi:10.7309/jmtm.6.3.5.
24. **Majbah Uddin** and Nathan Huynh. 2016. "Routing Model for Multicommodity Freight in an Intermodal Network under Disruptions." *Transportation Research Record* 2548, 71–80. doi:10.3141/2548-09.
25. **Majbah Uddin** and Nathan Huynh. 2015. "Freight Traffic Assignment Methodology for Large-Scale Road-Rail Intermodal Networks." *Transportation Research Record* 2477, 50–57. doi:10.3141/2477-06.
26. **Majbah Uddin**, Nathan Huynh, Jose Vidal, Kevin Taaffe, Lawrence Fredendall, and Joel Greenstein. 2015. "Evaluation of Google's Voice Recognition and Sentence Classification for Health Care Applications." *Engineering Management Journal* 27(3), 152–162. doi:10.1080/10429247.2015.1054752.

Manuscripts Under Review

1. Latif Patwary, Pankaj Dahal, **Majbah Uddin**, and Stacy C Davis. Under review. "Web Logs as Digital Footprints: Methods, Applications, and Research Gaps." *Library & Information Science Research*.
2. **Majbah Uddin**, Ingrid Busch, Nicholas Gregorich, Mariappan Parans Paranthaman, Ramesh Bhave, and Syed Z Islam. Under review. "Multimodal Transportation Optimization for Mining Waste Utilization in Ocean Alkalinity Enhancement: A Mixed-Integer Linear Programming Approach." *Journal of CO_2 Utilization*.
3. Melrose Pan, Pankaj Dahal, Latif Patwary, and **Majbah Uddin**. Under review. "Sequential Activity Patterns and On-Demand Transportation: A Transformer-Based Analysis of Daily Trip Chains." *Transportation Research Record*.
4. Latif Patwary, Xiaomin Chen, Md Sami Hasnine, and **Majbah Uddin**. Under review. "Spatiotemporal Modeling of E-Scooter Demand in a Mid-Sized US City Using a Bayesian Hierarchical Framework." *Transportation Research Record*.
5. Latif Patwary, Md Sami Hasnine, **Majbah Uddin**, and Asad J Khattak. Under review. "Emerging Challenges of EV Ownership: No-Home Charging and Induced Travel Demand." *International Journal of Sustainable Transportation*.
6. Nicholas Gregorich, Shailesh Dangwal, Stephanie Cho, Saurabh Prakash Pethe, Mary Hannah Irwin, Mustafa Abbas, Robert Sacci, **Majbah Uddin**, Benjamin T Manard, Mariappan Parans Paranthaman, Jeffrey R McCutcheon, Ramesh Bhave, and Syed Z Islam. Under review. "Critical Materials Waste-Enabled Ocean Alkalinity Enhancement." *Chemical Engineering Journal*.

Conference Papers

1. Latif Patwary, Md Sami Hasnine, and **Majbah Uddin**. 2025. "Are Baby Boomers' Non-Work Trip-Making Behaviors Different Than Millennials? Lessons Learned from NHTS Data." *Proceedings of the International Conference on Transportation and Development*, 51–63. doi:10.1061/9780784486207.005.
2. Melrose Pan, Christa Brelsford, and **Majbah Uddin**. 2024. "A Spatial-Temporal Analysis of Travel Time Gap and Inequality between Public Transportation and Personal Vehicles." *Proceedings of the International Conference on Transportation and Development*, 554–564. doi:10.1061/9780784485521.050.
3. **Majbah Uddin**, Yuandong Liu, and Hyeonsup Lim. 2023. "Exploring the Effects of Population and Employment Characteristics on Truck Flows: An Analysis of NextGen NHTS Origin-Destination Data." *Proceedings of the International Conference on Transportation and Development*, 503–513. doi:10.1061/9780784484883.044.
4. Yuandong Liu and **Majbah Uddin**. 2023. "Mobility Gaps between Low-Income and Not Low-Income Households: A Case Study in New York State." *Proceedings of the International Conference on Transportation and Development*, 663–673. doi:10.1061/9780784484883.057.
5. **Majbah Uddin**, Yuandong Liu, Fei Xie, Vivek Sujan, and Adam Siekmann. 2022. "Modeling Freight Traffic Demand and Highway Networks for Hydrogen Fueling Station Planning: A Case Study of US Interstate 75 Corridor." *Proceedings of the 14th International Green Energy Conference (IGEC 2022)*. OSTI:1875348.
6. Yuandong Liu, Adam Siekmann, Vivek Sujan, **Majbah Uddin**, Fei Xie, and Shiqi Ou. 2022. "Providing Levelized Cost and Waiting Time Inputs for HDV Hydrogen Refueling Station Planning: A Case Study of US I-75 Corridor." *Proceedings of the 14th International Green Energy Conference (IGEC 2022)*. OSTI:1876290.
7. Gurcan Comert, Charles Pierce, Zulfikar Berk, Nathan Huynh, Robert Petrulis, and **Majbah Uddin**. 2020. "Delivering Contextual Knowledge and Critical Skills of Disruptive Technologies through Problem-Based Learning in a Research Experiences for Undergraduates Setting." *Proceedings of the American Society for Engineering Education's Virtual Annual Conference*. doi:10.18260/1-2--34377.

Book Chapters

1. Nathan Huynh, **Majbah Uddin**, and Minh Cong Chu. 2025. "Data Analytics for Intermodal Freight Transportation Applications." M Chowdhury, K Dey, and A Apon, eds. *Data Analytics for Intelligent Transportation Systems*, 2nd edition, Chapter 10, 263–287. Elsevier. doi:10.1016/B978-0-443-13878-2.00013-8.
2. Nathan Huynh, **Majbah Uddin**, and Chu Cong Minh. 2017. "Data Analytics for Intermodal Freight Transportation Applications." M Chowdhury, A Apon, and K Dey, eds. *Data Analytics for Intelligent Transportation Systems*, 1st edition, Chapter 10, 241–262. Elsevier. doi:10.1016/B978-0-12-809715-1.00010-9.

Technical Reports

1. Hyeonsup Lim and **Majbah Uddin**. *Freight Analysis Framework Version 6 User's Guide*. Bureau of Transportation Statistics, US Department of Transportation, 2026. URL:bts.gov/faf.
2. Justin Jacobs, Aaron Myers, Annetta Burger, Jack Gonzales, Amy Moore, Steve Hespeler, Christopher Tate, Nicolle Di Domenico, and **Majbah Uddin**. *FIVEKAY TA-2 FY25 Technical Report: Reidentification Feasibility Study*. ORNL/SPR-2026/4606. Oak Ridge National Laboratory, 2026.

3. Latif Patwary, **Majbah Uddin**, Melrose Pan, and Hyeonsup Lim. *Exploring Nonmotorized Travel in New York State Using 2017 National Household Travel Survey Data*. ORNL/SPR-2025/3954. Report submitted to New York State Department of Transportation, 2025. doi:10.2172/3002356.
4. Melrose Pan, Hyeonsup Lim, **Majbah Uddin**, Latif Patwary, and Pankaj Dahal. *Analyzing Trip Chaining Behavior in New York State Using 2009 and 2017 National Household Travel Survey*. ORNL/SPR-2025/3955. Report submitted to New York State Department of Transportation, 2025. doi:10.2172/3002353.
5. Latif Patwary, Melrose Pan, and **Majbah Uddin**. *Travel Patterns and Characteristics of the Population from Zero Vehicle Households in New York State*. ORNL/SPR-2025/3768. Report submitted to New York State Department of Transportation, 2025. doi:10.2172/3002638.
6. Melrose Pan, Latif Patwary, Tim Reuscher, and **Majbah Uddin**. *New York State Household Travel Patterns: A Comprehensive Comparison Analysis*. ORNL/SPR-2025/3769. Report submitted to New York State Department of Transportation, 2025. doi:10.2172/3002637.
7. Chester Ford, Yuandong Liu, Alec Biehl, **Majbah Uddin**, Hyeonsup Lim, and Monique Stinson. *Development of the Freight Analysis Framework Version 5 (FAF5) Annual Estimates*. Bureau of Transportation Statistics, US Department of Transportation, 2025. URL:bts.gov/faf.
8. Hyeonsup Lim, Xuanke Wu, and **Majbah Uddin**. *Freight Flow Analysis in New York State: A Comparison of FAF and Transearch*. ORNL/SPR-2025/3767. Report submitted to New York State Department of Transportation, 2025.
9. Melrose Pan and **Majbah Uddin**. *A Pilot Study on County-Level Disaggregation of National Household Travel Survey Origin-Destination Data*. ORNL/SPR-2025/3761. Report submitted to New York State Department of Transportation, 2025.
10. Latif Patwary, **Majbah Uddin**, Pankaj Dahal, Melrose Pan, and Hyeonsup Lim. *Nonhighway Gasoline and Special Fuel Consumption Estimation Model Study Report*. ORNL/SPR-2024/3314. Report submitted to Federal Highway Administration, US Department of Transportation, 2025.
11. Latif Patwary, Melrose Pan, **Majbah Uddin**, and Tim Reuscher. *Travel Patterns and Characteristics of Millennial Population in New York State*. ORNL/SPR-2024/3297. Report submitted to New York State Department of Transportation, 2025. doi:10.2172/3002915.
12. Melrose Pan, Latif Patwary, and **Majbah Uddin**. *Origin-Destination Passenger and Freight Travel Flows in New York State: A Comparison of Next-Generation National Household Travel Survey, Replica, and Geotab Datasets*. ORNL/SPR-2024/3430. Report submitted to New York State Department of Transportation, 2024.
13. Melrose Pan, Latif Patwary, and **Majbah Uddin**. *Travel Patterns and Characteristics of Population in Rural Areas of New York State*. ORNL/SPR-2024/3216. Report submitted to New York State Department of Transportation, 2024. doi:10.2172/2394748.
14. **Majbah Uddin**, Melrose Pan, Yuandong Liu, and Tim Reuscher. *Travel Patterns and Characteristics of Low-Income Population in New York State: 2017 Update*. ORNL/TM-2023/2942. Report submitted to New York State Department of Transportation, 2024. doi:10.2172/2301664.
15. Melrose Pan and **Majbah Uddin**. *The Utility of National Household Travel Survey Add-on Data for New York State*. ORNL/SPR-2024/3215. Report submitted to New York State Department of Transportation, 2024.
16. Melrose Pan and **Majbah Uddin**. *A Comparison of Next-Generation National Household Travel Survey Origin-Destination Data and Replica Trip Data*. ORNL/TM-2023/2920. Report submitted to New York State Department of Transportation, 2023.
17. Melrose Pan, **Majbah Uddin**, Alec Biehl, and Ho-Ling Hwang. *Alternative Fuel Vehicle Usage and Owner Demographics in New York State*. ORNL/TM-2023/2865. Report submitted to New York State Department of Transportation, 2023. doi:10.2172/1996693.
18. **Majbah Uddin** and Hyeonsup Lim. *Freight Flows in Onondaga County, New York: A Comparison of Three Data Sources*. ORNL/TM-2022/2743. Report submitted to New York State Department of Transportation, 2023.
19. Hyeonsup Lim, **Majbah Uddin**, and Yuandong Liu. *Quick Validation Plan for FAF5 Truck Network Assignment*. ORNL/TM-2022/2804. Report submitted to US Department of Transportation, 2022.
20. **Majbah Uddin** and Ho-Ling Hwang. *Travel Statistics for New York State Department of Transportation Regions*. ORNL/TM-2022/2455. Report submitted to New York State Department of Transportation, 2022.
21. Yuandong Liu, Ho-Ling Hwang, **Majbah Uddin**, Tim Reuscher, and Shih-Miao Chin. *Travel Patterns and Characteristics of Elderly Population in New York State: 2017 Update*. ORNL/TM-2022/2614. Report submitted to New York State Department of Transportation, 2022. doi:10.2172/1899831.
22. US Department of Energy Office of Science. *US Department of Energy: National Virtual Biotechnology Laboratory Technical Report*. US Department of Energy, 2022. doi:10.2172/1871217. **Research Team Member**, Chapter 3: Epidemiological Modeling.
23. US Department of Energy Office of Science. *National Virtual Biotechnology Laboratory: Report on Rapid R&D Solutions to the COVID-19 Crisis*. US Department of Energy, 2021. doi:10.2172/1968213. **Research Team Member**, Epidemiological Modeling.
24. Ho-Ling Hwang, Hyeonsup Lim, Shih-Miao Chin, **Majbah Uddin**, Alec Biehl, Fei Xie, Stephanie Hargrove, Yuandong Liu, and Ross Wang. *Freight Analysis Framework Version 5 (FAF5) Base Year 2017 Data Development Technical Report*. ORNL/TM-2021/2154. Bureau of Transportation Statistics, US Department of Transportation, 2021. doi:10.2172/1844893. URL:bts.gov/faf.
25. Ho-Ling Hwang, Shih-Miao Chin, Hyeonsup Lim, Ross Wang, Fei Xie, **Majbah Uddin**, Alec Biehl, and Yuandong Liu. *Freight Analysis Framework Version 5 User's Guide*. Bureau of Transportation Statistics, US Department of Transportation, 2021. URL:bts.gov/faf.
26. Ho-Ling Hwang, Shih-Miao Chin, Hyeonsup Lim, **Majbah Uddin**, Ross Wang, Fei Xie, Alec Biehl, and Yuandong Liu. *Freight Analysis Framework Version 5 Getting Started Guide for the Data Tabulation Tool (DTT)*. Bureau of Transportation Statistics, US Department of Transportation, 2021. URL:bts.gov/faf.
27. **Majbah Uddin**, Ho-Ling Hwang, Yuandong Liu, and Tim Reuscher. *County-Level Travel Statistics for Capital District Transportation Committee*.

ORNL/TM-2021/2097. Report submitted to New York State Department of Transportation, 2021.

28. Ho-Ling Hwang, Tim Reuscher, and **Majbah Uddin**. *Emission Factors Profile for New York State: Travel Statistics for Regional Air Quality Modeling, based on 2017 National Household Travel Survey Data*. ORNL/TM-2021/2093. Report submitted to New York State Department of Transportation, 2021.
29. **Majbah Uddin**, Yuandong Liu, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: New York City Regional Economic Development Council*. ORNL/TM-2021/1936. Report submitted to New York State Department of Transportation, 2021.
30. **Majbah Uddin**, Yuandong Liu, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: Mohawk Valley Regional Economic Development Council*. ORNL/TM-2021/1953. Report submitted to New York State Department of Transportation, 2021.
31. **Majbah Uddin**, Yuandong Liu, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: Mid-Hudson Regional Economic Development Council*. ORNL/TM-2021/1952. Report submitted to New York State Department of Transportation, 2021.
32. **Majbah Uddin**, Yuandong Liu, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: Long Island Regional Economic Development Council*. ORNL/TM-2021/1951. Report submitted to New York State Department of Transportation, 2021.
33. **Majbah Uddin**, Yuandong Liu, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: Finger Lakes Regional Economic Development Council*. ORNL/TM-2021/1950. Report submitted to New York State Department of Transportation, 2021.
34. **Majbah Uddin**, Yuandong Liu, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: Central New York Regional Economic Development Council*. ORNL/TM-2021/1949. Report submitted to New York State Department of Transportation, 2021.
35. **Majbah Uddin**, Yuandong Liu, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: Capital Region Regional Economic Development Council*. ORNL/TM-2021/1948. Report submitted to New York State Department of Transportation, 2021.
36. Yuandong Liu, **Majbah Uddin**, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: North Country Regional Economic Development Council*. ORNL/TM-2021/1954. Report submitted to New York State Department of Transportation, 2021.
37. Yuandong Liu, **Majbah Uddin**, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: Southern Tier Regional Economic Development Council*. ORNL/TM-2021/1955. Report submitted to New York State Department of Transportation, 2021.
38. Yuandong Liu, **Majbah Uddin**, Ho-Ling Hwang, and Shih-Miao Chin. *Summary of 2017 National Household Travel Survey Statistics: Western New York Regional Economic Development Council*. ORNL/TM-2021/1956. Report submitted to New York State Department of Transportation, 2021.
39. Yuandong Liu, **Majbah Uddin**, Shih-Miao Chin, Ho-Ling Hwang, and Hyeonsup Lim. *Data Sources and Methodologies for Compiling Pipeline Intermodal Terminal Dataset*. ORNL/TM-2021/2007. Report submitted to Bureau of Transportation Statistics, US Department of Transportation, 2021.
40. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Watertown Jefferson County Area Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1690. Report submitted to New York State Department of Transportation, 2020.
41. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Herkimer-Oneida Counties Transportation Study: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1689. Report submitted to New York State Department of Transportation, 2020.
42. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Syracuse Metropolitan Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1688. Report submitted to New York State Department of Transportation, 2020.
43. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Statewide: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1687. Report submitted to New York State Department of Transportation, 2020.
44. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Small Urban Areas: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1686. Report submitted to New York State Department of Transportation, 2020.
45. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Small Rural Areas: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1685. Report submitted to New York State Department of Transportation, 2020.
46. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Genesee Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1684. Report submitted to New York State Department of Transportation, 2020.
47. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Putnam, Rockland, Westchester: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1683. Report submitted to New York State Department of Transportation, 2020.
48. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Dutchess County Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1682. Report submitted to New York State Department of Transportation, 2020.
49. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *New York Metropolitan Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1681. Report submitted to New York State Department of Transportation, 2020.
50. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *New York City – Bronx, Kings, Queens, New York, Richmond: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1680. Report submitted to New York State Department of Transportation, 2020.
51. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Orange County Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1679. Report submitted to New York State Department of Transportation, 2020.

52. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Nassau, Suffolk: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1678. Report submitted to New York State Department of Transportation, 2020.
53. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Manhattan: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1677. Report submitted to New York State Department of Transportation, 2020.
54. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Ulster County Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1676. Report submitted to New York State Department of Transportation, 2020.
55. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Ithaca-Tompkins County Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1675. Report submitted to New York State Department of Transportation, 2020.
56. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Adirondack-Glens Falls Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1674. Report submitted to New York State Department of Transportation, 2020.
57. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Elmira-Chemung Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1673. Report submitted to New York State Department of Transportation, 2020.
58. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Greater Buffalo-Niagara Regional Transportation Council: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1672. Report submitted to New York State Department of Transportation, 2020.
59. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Binghamton Metropolitan Transportation Study: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1667. Report submitted to New York State Department of Transportation, 2020.
60. Ho-Ling Hwang, Tim Reuscher, Yuandong Liu, **Majbah Uddin**, and Alec Biehl. *Capital District Transportation Committee: 2017 National Household Travel Survey, New York Add-On*. ORNL/TM-2020/1665. Report submitted to New York State Department of Transportation, 2020.
61. Nathan Huynh, **Majbah Uddin**, and Samaneh Shiri. *Evaluation of Jamar Black Cat Radar Recorders*. Report submitted to South Carolina Department of Transportation, 2017.

Datasets

1. US Department of Transportation, Bureau of Transportation Statistics (BTS). 2026. *Freight Analysis Framework, FAF6* [datasets]. doi:10.21949/fhgm-g335. **Development Team Member**
2. US Department of Transportation, Bureau of Transportation Statistics (BTS). 2021. *Intermodal Freight Facilities Pipeline Terminals* [dataset]. National Transportation Atlas Database (NTAD). URL:geodata.bts.gov. **Development Team Member**
3. US Department of Transportation, Bureau of Transportation Statistics (BTS), Federal Highway Administration (FHWA). 2021. *Freight Analysis Framework, FAF5* [datasets], 2017–2050 data. doi:10.21949/1529116. **Development Team Member**
4. US Department of Transportation, Bureau of Transportation Statistics (BTS), Federal Highway Administration (FHWA). 2021. *Freight Analysis Framework, FAF5* [datasets], historical reprocessed 1997–2012 data. doi:10.21949/1529108. **Development Team Member**

In addition, developed 20+ sponsor-delivered and internal datasets (not publicly released) for FHWA, BTS, NYSDOT, DOE, and ORNL LDRD.

Software Tools

1. Hyeonsup Lim, Pankaj Dahal, and **Majbah Uddin**. 2026. "FAF6 Data Tabulation Tool (DTT)." Developed for the Bureau of Transportation Statistics. URL:bts.gov/faf.
2. **Majbah Uddin** and Pankaj Dahal. 2026. "Fueleconomy.gov Traffic Report Generator." Developed under the Fuel Economy Guide project sponsored by DOE.
3. Pankaj Dahal and **Majbah Uddin**. 2026. "FEG Web Traffic Analytics." Developed under the Fuel Economy Guide project sponsored by DOE.
4. Pankaj Dahal and **Majbah Uddin**. 2026. "New York State Community Profiles Prototype Tool." Developed for the New York State Department of Transportation.
5. Hyeonsup Lim, Qixing Wang, and **Majbah Uddin**. 2026. "NYSDOT Freight Flow Analysis Tool." Developed for the New York State Department of Transportation.
6. Latif Patwary, **Majbah Uddin**, and Pankaj Dahal. 2025. "Nonhighway Gasoline and Diesel Consumption Estimation Tool." Developed for the Federal Highway Administration.
7. Robin Clark and **Majbah Uddin**. 2024. "CCUS-Agent: Agent-Based Simulation Model for Analyzing Multimodal Transportation of CO_2 for CCUS." doi:10.11578/dc.20240129.2.

Software Copyright

1. Pankaj Dahal, **Majbah Uddin**, and Stacy C Davis. 2026. "Immutable-Profile Web Analytics System for Auditable Replay of Server-Log Traffic Metrics." *Site Accession No. 90000504*. DOE CODE:183385.

2. Robin Clark and **Majbah Uddin**. 2024. "Agent-Based Simulation Model for Analyzing Multimodal Transportation of CO_2 for CCUS." *US Copyright Registration No. TXu 2-413-830*. doi:10.11578/dc.20240129.2.

Invited Talks and Panels

1. **Majbah Uddin**. 2026. "Understanding Freight and Passenger Travel Through Data Science." *Utah State University Data Science Club*, Virtual.
2. **Majbah Uddin**. 2026. "Advanced Forecasting and Optimization." *DOE Transportation Technologies Office (TTO) Analysis AI Workshop*, Washington, DC.
3. **Majbah Uddin**. 2026. **Panelist**, session on "Inefficiencies and Supply." *ORNL Collaboration Catalysts: Secure, Resilient Systems for Abundant Energy*, Oak Ridge, TN.
4. **Majbah Uddin**. 2026. "FAF and NHTS: National Freight and Passenger Travel Data for Regional Resilience Analysis." *Accelerating Southeast Resilience Workshop*, Oak Ridge, TN.
5. **Majbah Uddin**. 2026. "Technology-Enabled Approaches for Mobility Management." *Workshop on Evaluating Mobility Management Tools and Strategies for Implementation and Practice, Transportation Research Board Annual Meeting*, Washington, DC.
6. **Majbah Uddin**. 2025. "Transportation Analytics and Decision Sciences." *German-American Frontiers of Engineering Symposium, National Academy of Engineering*, Oak Ridge, TN.
7. **Majbah Uddin** and Richard Batchelder. 2025. "National Household Travel Survey." *NYSHAReD Community of Practice*, Virtual.
8. **Majbah Uddin**. 2019. "Assessing Service Life of Traffic Signs in South Carolina." *ASCE Midlands Branch Meeting*, Columbia, SC.

Presentations

1. Qixing Wang, Pankaj Dahal, Jaesup Lee, and **Majbah Uddin**. "National Freight Flow Modeling in the Big Data and AI Era: A Review of Four-Step, Agent-Based, and Data-Driven Paradigms." *Transportation Research Board Annual Meeting*, Washington, DC, January 2027.
2. **Majbah Uddin**, Xuanke Wu, Latif Patwary, Hyeonsup Lim, and Monique Stinson. "Estimating Regional Crude Petroleum and Natural Gas Freight Flows in the United States: Data Fusion Lessons from the Freight Analysis Framework Version 6." *Transportation Research Board Annual Meeting*, Washington, DC, January 2027.
3. Melrose Pan, Pankaj Dahal, Latif Patwary, and **Majbah Uddin**. "Sequential Activity Patterns and On-Demand Transportation: A Transformer-Based Analysis of Daily Trip Chains." *Transportation Research Board Annual Meeting*, Washington, DC, January 2027.
4. Nicholas Gregorich, Joseph Kalla, Mary Hannah Irwin, Saurabh Prakash Pethe, Mustafa Abbas, Zachary Coin, Mariappan Parans Paranthaman, **Majbah Uddin**, Shailesh Dangwal, Ramesh Bhave, and Syed Z Islam. "Mining Waste for Energy Efficient Carbon Capture via Ocean Alkalinity Enhancement." *AIChE Annual Meeting*, Minneapolis, MN, November 2026.
5. Pankaj Dahal, **Majbah Uddin**, and Stacy C Davis. "Immutable-Profile Web Analytics System for Auditable Replay of Server-Log Traffic Metrics." *ORNL Software and Data Expo*, Oak Ridge, TN, August 2026.
6. Latif Patwary, Xiaomin Chen, Md Sami Hasnine, and **Majbah Uddin**. "Understanding Spatially Varying Determinants of E-Scooter Ridership for Smarter Urban Mobility." *Oak Ridge Postdoctoral Association Research Symposium*, Oak Ridge, TN, May 2026.
7. Latif Patwary, Xiaomin Chen, Md Sami Hasnine, and **Majbah Uddin**. "Spatio-Temporal Modeling of E-Scooter Demand in a Mid-Sized US City Using a Bayesian Hierarchical Framework." *Transportation Research Board Annual Meeting*, Washington, DC, January 2026.
8. Qixing Wang, Hyeonsup Lim, Xuanke Wu, and **Majbah Uddin**. "Game-Theoretic Min-Max Framework for Large-Scale Freight Network Vulnerability Assessment under Adversarial Disruptions." *Transportation Research Board Annual Meeting*, Washington, DC, January 2026.
9. Latif Patwary and **Majbah Uddin**. "A Tool for Nonhighway Gasoline and Diesel Fuel Consumption Estimation in the US." *ORNL Software and Data Expo*, Oak Ridge, TN, September 2025.
10. Latif Patwary, Melrose Pan, **Majbah Uddin**, and Tim Reuscher. "Travel Patterns and Characteristics of Millennial Population in New York." *Oak Ridge Postdoctoral Association Research Symposium*, Oak Ridge, TN, July 2025. **People's Choice Award**
11. Latif Patwary, Md Sami Hasnine, and **Majbah Uddin**. "Are Baby Boomers' Non-Work Trip-Making Behavior Different than Millennials? Lessons Learned from NHTS Data." *International Conference on Transportation and Development*, Glendale, AZ, June 2025.
12. Latif Patwary and **Majbah Uddin**. "Data-Driven Insights for New York State's MPOs: Travel Behavior Analysis Based on the NHTS Data." *New York State Association of Metropolitan Planning Organizations (NYSAMPO) Conference*, Syracuse, NY, May 2025.
13. Melrose Pan, Latif Patwary, and **Majbah Uddin**. "Public Transportation Usage among Night Commuters: The Role of Land Use and Accessibility." *Transportation Research Board Annual Meeting*, Washington, DC, January 2025.
14. Latif Patwary, Melrose Pan, and **Majbah Uddin**. "An Equity Analysis on the Use of Public Transportation as an Alternative Mode in Rural Areas." *Transportation Research Board Annual Meeting*, Washington, DC, January 2025.
15. Latif Patwary, Md Sami Hasnine, **Majbah Uddin**, and Asad J Khattak. "Investigating the Emerging Challenges of Electric Vehicle Ownership: No-Home Charging Infrastructure and the Case of Induced Travel Demand." *Transportation Research Board Annual Meeting*, Washington, DC, January 2025.

16. Latif Patwary and **Majbah Uddin**. "Investigating the Endogeneity of Electric Vehicle Adoption and Charging Infrastructure: Evidence from the State of Washington." *Transportation Research Board Annual Meeting*, Washington, DC, January 2025.
17. Amy Moore, Hyeonsup Lim, **Majbah Uddin**, Pankaj Dahal, Latif Patwary, Melrose Pan, Michael Sharp, and Michael Kass. "Who Favors 'Green' Shipping?" *Transportation Research Board Annual Meeting*, Washington, DC, January 2025.
18. Jiafu Mao, Yaoping Wang, Fengming Yuan, **Majbah Uddin**, Xin Gu, Daniel M Ricciuto, Xiaoying Shi, Ingrid Busch, Dan Lu, Nagendra Singh, Peter E Thornton, David L McCollum, Andrew G Stack, and David Sholl. "A Combined Ecosystem and Engineering Decision-Making Framework for Optimized Enhanced Weathering." *AGU24 Annual Meeting*, Washington, DC, December 2024.
19. Latif Patwary, **Majbah Uddin**, Pankaj Dahal, Melrose Pan, and Hyeonsup Lim. "Estimating Gasoline Consumption for the Recreational Boating Sector in the US" *Oak Ridge Postdoctoral Association Research Symposium*, Oak Ridge, TN, July 2024.
20. Melrose Pan, Christa Brelsford, and **Majbah Uddin**. "A Spatial-Temporal Analysis of Travel Time Gap and Inequality between Public Transportation and Personal Vehicles." *International Conference on Transportation and Development*, Atlanta, GA, June 2024.
21. **Majbah Uddin** and Robin Clark. "Multimodal Transportation of CO_2 to Support Large-Scale Carbon Capture, Utilization and Storage." *Innovations in Climate Resilience Conference*, Washington, DC, April 2024.
22. Diyi Liu, Hyeonsup Lim, **Majbah Uddin**, Yuandong Liu, Lee D Han, Ho-Ling Hwang, and Shih-Miao Chin. "Improving the Accuracy of Freight Mode Choice Models: A Case Study Using the 2017 Commodity Flow Survey (CFS) Public Use File Data Set and Ensemble Learning Techniques." *Transportation Research Board Annual Meeting*, Washington, DC, January 2024.
23. Melrose Pan and **Majbah Uddin**. "Temporal Evolution of Travel Time Gap between Public Transit and Personal Vehicle and Their Inequality: An Analysis Using Census Transportation Planning Products Program Data." *Transportation Research Board Annual Meeting*, Washington, DC, January 2024.
24. **Majbah Uddin**, Melrose Pan, and Ho-Ling Hwang. "Unraveling Factors Influencing Mode Choice of Adults with Travel-Limiting Disabilities Using Random Parameters Logit Modeling." *Transportation Research Board Annual Meeting*, Washington, DC, January 2024.
25. Melrose Pan, **Majbah Uddin**, and Hyeonsup Lim. "Evaluating Geospatial Patterns of Electric Vehicle Ownership Using Multi-Scale Geographically Weighted Method." *Transportation Research Board Annual Meeting*, Washington, DC, January 2024.
26. Robin Clark and **Majbah Uddin**. "An Agent-Based Tool for Carbon Capture, Utilization and Storage Transportation Flows and Infrastructure." *ORNL Software and Data Expo*, Oak Ridge, TN, September 2023.
27. **Majbah Uddin**, Robin Clark, Michael Hilliard, Josh Thompson, Matthew Langholtz, and Erin Webb. "Modeling a Gigaton CO_2 Economy." *ORNL LDRD Poster Fair*, Oak Ridge, TN, September 2023.
28. Diyi Liu, Hyeonsup Lim, Vivek Sujan, Adam Siekmann, Amy Moore, **Majbah Uddin**, and Ankur Shiledar. "A Heuristic Method to Sequentially Impute Vehicle Class and Weight for US Highway Network Truck Flows." *TRB Innovations in Freight Data Workshop*, Washington, DC, September 2023.
29. **Majbah Uddin**, Ho-Ling Hwang, and Md Sami Hasnine. "Understanding Bikeshare Trip Demand in New York City: An Application of Interpretable Machine Learning Framework." *ORNL AI Expo*, Oak Ridge, TN, September 2023.
30. **Majbah Uddin**, Yuandong Liu, and Hyeonsup Lim. "Exploring the Effects of Population and Employment Characteristics on Truck Flows: An Analysis of NextGen NHTS Origin-Destination Data." *International Conference on Transportation and Development*, Austin, TX, June 2023.
31. Yuandong Liu and **Majbah Uddin**. "Mobility Gaps between Low-Income and Not Low-Income Households: A Case Study in New York State." *International Conference on Transportation and Development*, Austin, TX, June 2023.
32. Melrose Pan and **Majbah Uddin**. "Identifying Travel Patterns of Low-Income Populations in New York State." *Oak Ridge Postdoctoral Association Research Symposium*, Oak Ridge, TN, June 2023.
33. Robin Clark, **Majbah Uddin**, Michael Hilliard, Josh Thompson, Matthew Langholtz, and Erin Webb. "Modeling Multimodal CO_2 Transportation for Carbon Capture, Utilization, and Storage." *Annual Modeling and Simulation Conference*, Hamilton, ON, Canada, May 2023.
34. Yuandong Liu, **Majbah Uddin**, Shih-Miao Chin, Ho-Ling Hwang, and Jiaoli Chen. "Examining Rail Transportation Route of Crude Oil in the US Using Crowdsourced Social Media Data." *Transportation Research Board Annual Meeting*, Washington, DC, January 2023.
35. Hyeonsup Lim, **Majbah Uddin**, Yuandong Liu, Shih-Miao Chin, and Ho-Ling Hwang. "Multi-Output Chaining for Time Series Forecasting to Estimate Impact of Supply Chain Disruption: The Case of Russia's Invasion of Ukraine on US Imports and Exports." *Transportation Research Board Annual Meeting*, Washington, DC, January 2023.
36. Yuandong Liu, **Majbah Uddin**, Ho-Ling Hwang, and Shih-Miao Chin. "Exploring the Transportation Mobility Disparities between Elderly and Non-Elderly Population in the State of New York Using Census Transportation Planning Products and National Household Travel Survey Data." *Transportation Research Board Annual Meeting*, Washington, DC, January 2023.
37. **Majbah Uddin**, Ho-Ling Hwang, and Md Sami Hasnine. "An Interpretable Machine Learning Framework to Understand Bikeshare Demand before and during the COVID-19 Pandemic in New York City." *Transportation Research Board Annual Meeting*, Washington, DC, January 2023.
38. **Majbah Uddin**, Ho-Ling Hwang, and Shih-Miao Chin. "Sentiment Analytics as Part of Public Participation in the Transportation Scenario Planning." *TRB Conference on Scenario Planning in Transportation*, Washington, DC, September 2022.
39. Yuandong Liu, **Majbah Uddin**, Ho-Ling Hwang, and Tim Reuscher. "Travel Patterns and Characteristics of Elderly Population in the State of New York." *Oak Ridge Postdoctoral Association Research Symposium*, Oak Ridge, TN, July 2022. **Poster Presentation Award**

40. **Majbah Uddin**, Yuandong Liu, Fei Xie, Vivek Sujan, and Adam Siekmann. "Modeling Freight Traffic Demand and Highway Networks for Hydrogen Fueling Station Planning: A Case Study of US Interstate 75 Corridor." *14th International Green Energy Conference (IGEC-XIV)*, Virtual, July 2022.
41. Yuandong Liu, Adam Siekmann, Vivek Sujan, **Majbah Uddin**, Fei Xie, and Shiqi Ou. "Providing Levelized Cost and Waiting Time Inputs for HDV Hydrogen Refueling Station Planning: A Case Study of US I-75 Corridor." *14th International Green Energy Conference (IGEC-XIV)*, Virtual, July 2022.
42. Hyeonsup Lim, **Majbah Uddin**, Yuandong Liu, Shih-Miao Chin, and Ho-Ling Hwang. "A Comparative Study of Machine Learning Algorithms for an Industry-Specific Freight Generation Model." *Transportation Research Board Annual Meeting*, Washington, DC, January 2022.
43. **Majbah Uddin**, Ho-Ling Hwang, and Shih-Miao Chin. "Evaluating Public Perception towards Autonomous Goods Delivery Services Using Social Media Data." *TRB Innovations in Freight Data Workshop*, Virtual, September 2021.
44. Yuandong Liu, **Majbah Uddin**, Shih-Miao Chin, and Ho-Ling Hwang. "Estimation of Crude Oil Movements for the Freight Analysis Framework Using Data Fusion." *TRB Innovations in Freight Data Workshop*, Virtual, September 2021.
45. Hyeonsup Lim, Chieh Wang, Ho-Ling Hwang, Shih-Miao Chin, and **Majbah Uddin**. "A Two-Step Approach for Truck Load Factor Estimation Based on the Travel Monitoring Analysis System Data Using K-Means Clustering and Gaussian Mixture Model." *TRB Innovations in Freight Data Workshop*, Virtual, September 2021.
46. **Majbah Uddin** and Ho-Ling Hwang. "Access to Public Transportation in Rural New York State: An Equity Analysis Using 2017 NHTS Data." *TRB Conference on Advancing Transportation Equity*, Virtual, September 2021.
47. Michael Hilliard, Alec Biehl, Ho-Ling Hwang, Byung Park, Chieh Wang, Fei Xie, and **Majbah Uddin**. "Evaluating Inequitable Disruptions to Commuters Using a Synthetic Agent-Based Simulation: A Case Study of the COVID-19 Pandemic in New York City." *TRB Conference on Advancing Transportation Equity*, Virtual, September 2021.
48. Alec Biehl, **Majbah Uddin**, and Ho-Ling Hwang. "Assessing Transportation Equity in Rural New York Counties." *TRB Conference on Advancing Transportation Equity*, Virtual, September 2021.
49. **Majbah Uddin**, Yuandong Liu, and Ho-Ling Hwang. "Visualizing Travel Patterns in New York State Regional Economic Development Council Regions." *Oak Ridge Postdoctoral Association Research Symposium*, Virtual, July 2021.
50. Yuandong Liu, **Majbah Uddin**, Shih-Miao Chin, Ho-Ling Hwang, and Hyeonsup Lim. "Compilation of the US Pipeline Intermodal Terminal Dataset." *Oak Ridge Postdoctoral Association Research Symposium*, Virtual, July 2021.
51. **Majbah Uddin** and Ho-Ling Hwang. "Factors Influencing Mode Choice of Adults with Travel-Limiting Disability." *International Conference on Transportation and Development*, Virtual, June 2021.
52. **Majbah Uddin**, Shih-Miao Chin, Ho-Ling Hwang, Chieh Wang, and Hyeonsup Lim. "Assessing the Potential Impact of Lane-Tracking Technology for Autonomous Vehicles on Traffic Safety." *International Conference on Transportation and Development*, Virtual, June 2021.
53. Hyeonsup Lim, Yuandong Liu, **Majbah Uddin**, Ho-Ling Hwang, and Shih-Miao Chin. "Development of US Nationwide Freight Flow and Infrastructure Data." *ORNL Software and Data Expo*, Virtual, May 2021. **Best Demo Award**
54. **Majbah Uddin** and Ho-Ling Hwang. "A Bi-Objective Optimization Framework for Municipal Solid Waste Transportation." *INFORMS Annual Meeting*, Virtual, November 2020.
55. **Majbah Uddin** and Shih-Miao Chin. "Movements of Construction and Demolition Debris in the United States." *Annual Vanderbilt Postdoctoral Association Symposium*, Virtual, October 2020.
56. **Majbah Uddin**, Ho-Ling Hwang, and Stephanie Hargrove. "Movements of Municipal Solid Waste in the United States." *Oak Ridge Postdoctoral Association Research Symposium*, Virtual, July 2020.
57. **Majbah Uddin**, Shih-Miao Chin, and Ho-Ling Hwang. "Multimodal Transportation Flows of Crude Oil in the United States." *Oak Ridge Postdoctoral Association Research Symposium*, Virtual, July 2020.
58. **Majbah Uddin**, Sabreena Anowar, and Naveen Eluru. "Modeling Freight Mode Choice Using Machine Learning Classifiers: A Comparative Study Using the Commodity Flow Survey Data." *Transportation Research Board Annual Meeting*, Washington, DC, January 2020.
59. **Majbah Uddin** and Nathan Huynh. "Injury Severity Analysis of Truck-Involved Crashes under Different Weather Conditions." *Transportation Research Board Annual Meeting*, Washington, DC, January 2020.
60. **Majbah Uddin** and Nathan Huynh. "Model for Collaboration among Carriers in Domestic Intermodal Freight Transport." *Transportation Research Board Annual Meeting*, Washington, DC, January 2020.
61. **Majbah Uddin**, Nathan Huynh, and Fahim Ahmed. "Assignment of Freight Traffic in a Large-Scale Intermodal Network under Uncertainty." *Transportation Research Board Annual Meeting*, Washington, DC, January 2019.
62. **Majbah Uddin**. "Road-Rail Intermodal Freight Transport under Network Uncertainty: Development of Routing Models and Estimation of Environmental Impact." *VREF Advanced Studies Institutes on Sustainable Urban Freight Systems*, Troy, NY, August 2018.
63. **Majbah Uddin** and Nathan Huynh. "Assessing the Impact of Collaboration among Truck Carriers in Intermodal Freight Transportation." *5th Biennial Marine Transportation System Research & Development Conference*, Washington, DC, June 2018. **Student Honor Panel Presentation**
64. **Majbah Uddin**, Nathan Huynh, Lawrence Fredendall, and Kevin Taaffe. "Visualizing the Flow of Patients on the Day of Surgery." *POMS Annual Conference*, Houston, TX, May 2018.
65. **Majbah Uddin** and Nathan Huynh. "Reliable Routing of Road-Rail Intermodal Freight under Uncertainty." *Transportation Research Board*

Annual Meeting, Washington, DC, January 2018.

66. **Majbah Uddin** and Nathan Huynh. "Factors Influencing Injury Severity of Crashes Involving HAZMAT Trucks." *Transportation Research Board Annual Meeting*, Washington, DC, January 2018.
67. **Majbah Uddin** and Fahim Ahmed. "Analysis of Pedestrian Injury Severity in Motor Vehicle Crashes in Ohio." *Transportation Research Board Annual Meeting*, Washington, DC, January 2018.
68. **Majbah Uddin** and Nathan Huynh. "Truck-Involved Crashes Injury Severity Analysis for Different Lighting Conditions on Rural and Urban Roadways." *Transportation Research Board Annual Meeting*, Washington, DC, January 2017.
69. **Majbah Uddin** and Nathan Huynh. "Reliable Routing of Multicommodity Road-Rail Intermodal Freight under Uncertainty." *INFORMS Annual Meeting*, Nashville, TN, November 2016.
70. **Majbah Uddin**, Nathan Huynh, Jose Vidal, Kevin Taaffe, Lawrence Fredendall, and Joel Greenstein. "Evaluation of Google's Voice Recognition and Sentence Classification for Health Care Applications." *INFORMS Annual Meeting*, Nashville, TN, November 2016. **Poster Competition Semi-Finalist**
71. **Majbah Uddin** and Nathan Huynh. "A Routing Model for Multicommodity Freight in an Intermodal Network under Disruptions." *INFORMS Annual Meeting*, Philadelphia, PA, November 2015.
72. **Majbah Uddin** and Nathan Huynh. "Freight Traffic Assignment Methodology for Large-Scale Road-Rail Intermodal Networks." *Transportation Research Board Annual Meeting*, Washington, DC, January 2015.

Media Coverage

2024	**Elevated to IEEE Senior Member,** ORNL News article
2023	**Safe Train Transport,** ORNL News article
2023	**Photo-sharing website powers train disaster maps,** WVLT 8 feature
2023	**ORNL uses geotagged to recreate train routes to help with incidents,** WATE 6 feature
2022	**Transportation Influencer**, *TR News* No. 338, p. 46
2018	**Hazmat Crash Factors Identified,** *TR News* No. 316, News Briefs

Mentoring

2026–Present	**Qixing (Jason) Wang,** Postdoctoral Research Associate
2025–Present	**Pankaj Dahal,** Technical Professional Staff
July 2024–June 2025	**Michael Han,** GRO Student
May–August 2024	**Cameron James Swapp,** DOE SULI Internship
May–August 2024	**Samuel Arhin,** DOE SULI Internship
February 2024–Present	**A Latif Patwary,** Postdoctoral Research Associate
February 2023–November 2024	**Meiyu (Melrose) Pan,** Postdoctoral Research Associate
May–August 2022	**Eric Orozco,** GEM Fellow Internship
May–August 2022	**Thomas Vance,** DOE SULI Internship

Editorial and Advisory Roles

Handling Editor	*Transportation Research Record* (2026–Present)
Review Editor	*Frontiers in Future Transportation*
Panel Member	NCHRP 08-146, *Integrating Resiliency into Transportation System Operations* (2022–2024)
Working Group Member	TRB AR013 Working Group on Rail Workforce Development (2026–Present)
Program Committee	*UrbanAI'24: 2nd ACM SIGSPATIAL International Workshop on Advances in Urban-AI* (2024)

Peer Review

Reviewed 200+ papers for 20+ journals and conferences

Proposal Reviewer	*USDOT University Transportation Center R-SEAT, Florida A&M University–Florida State University*
Journal Referee	*Scientific Reports*, *Transportation Research Record*, *IEEE Access*, *Journal of Advanced Transportation*, *Sustainability*,

Conference Referee *Accident Analysis & Prevention* (Outstanding Reviewer in 2018), *Journal of Safety Research*, *Traffic Injury Prevention*, *International Journal of Environmental Research and Public Health*, *Applied Sciences*, *Information*, *PeerJ*, *Vehicles*, *International Journal of Injury Control and Safety Promotion*, *International Journal of Pavement Engineering*, *International Journal of Disaster Risk Reduction*, *IEEE Transactions on Intelligent Transportation Systems*, *International Journal of Urban Sciences*, *Transportation Research Part C*, *Transportation Research Part D*, *International Journal of Systems Science: Operations & Logistics*, *Transportation Planning and Technology*, *Transportmetrica A: Transport Science*, *Journal of Traffic and Transportation Engineering*, *Infrastructures*
Transportation Research Board Annual Meeting, *International Conference on Transportation and Development*, *World Conference on Transport Research*

Professional Services

2023–2025 **Session Chair,** Committee Meeting, Freight Systems and Marine Group Young Members Council, TRB Annual Meeting (3 times)
2023–2024 **Session Chair,** Freight Systems and Marine Transportation Work in Progress: A Lectern-Poster Session, TRB Annual Meeting (2 times)
2023 **Session Organizer,** The Future of Freight Panel, ORNL Smoky Mountains Mobility Conference, Chattanooga, TN
2016–2018 **Vice President,** Institute of Transportation Engineers, University of South Carolina Student Chapter
2015 **Member,** University of South Carolina Civil Engineering Department, Ad Hoc Committee for Enhanced Doctoral Experiences

Laboratory Service

2025 **Proposal Reviewer,** ORNL SEED/LDRD Program
2024 **Organizer,** Laboratory tours for staff of the Vehicle and Mobility Systems Research Section, ORNL

Professional Affiliations

2025–Present **Member,** TRB Standing Committee on Freight Rail Transportation and Supply Chain Reliability
2024–Present **Senior Member,** Institute of Electrical and Electronics Engineers (IEEE)
2023–2025 **Member,** ASCE T&DI Freight & Logistics Committee
2023–2025 **Corresponding Member,** ASCE T&DI Artificial Intelligence in Transportation Committee
2023–2025 **Member,** TRB Standing Committee on Agriculture and Food Transportation
2021–2025 **Chair,** TRB Freight Systems and Marine Group Young Members Council
2021–2025 **Member,** TRB Standing Committee on Freight Transportation Data
2021–2025 **Member,** TRB Subcommittee on Port Performance
2020–2025 **Member,** TRB Joint Subcommittee on Supply Chain Resilience
2020–2024 **Member,** TRB Standing Committee on Intermodal Freight Transport
2020–2022 **Member,** TRB Joint Subcommittee on Intermodal Freight Terminal Design and Operations
2019–Present **Associate Member,** American Society of Civil Engineers (ASCE)
2016–2023 **Member,** Institute for Operations Research and the Management Sciences (INFORMS)
2016–2019 **Member,** TRB Standing Committee on Intermodal Freight Terminal Design and Operations

Honors and Awards

2026 **Most Innovative Project Award,** ORNL Laboratory Directed Research and Development (LDRD) Poster Fair
2025 **Invited Participant,** German-American Frontiers of Engineering Symposium, National Academy of Engineering (NAE)
2025 **People's Choice Award,** Oak Ridge Postdoctoral Association Research Symposium
2024 **Innovation Award,** ORNL
2024 **Core Values Nomination for Teamwork,** ORNL
2023 **Core Values Nomination for Service,** ORNL
2022 **Poster Presentation Award,** Oak Ridge Postdoctoral Association Research Symposium

2021 **Best Demo Award,** ORNL Software and Data Expo

2020 **Most Cited Article Award,** *International Journal of Transportation Science and Technology*

2020 **Best Paper Award,** *Safety*

2019 **Breakthrough Graduate Scholar Award,** University of South Carolina

2016–2019 **M. Bert Storey Endowed Graduate Fellowship (4 times),** University of South Carolina

2018 **Participant,** Volvo Research and Educational Foundations Advanced Studies Institutes on Sustainable Urban Freight Systems, Rensselaer Polytechnic Institute

2018 **Student Honor Panel Presentation,** 5th Biennial Marine Transportation System Research & Development Conference

2018 **Preparing Future Faculty Award,** Center for Teaching Excellence, University of South Carolina

2017 **Highway Safety Information System Excellence in Highway Safety Data Award,** Federal Highway Administration

2016 **Highway Safety Data Research Paper Competition Award,** Federal Highway Administration

2015–2016 **Graduate School Travel Grant (2 times),** University of South Carolina

2015 **R. L. Sumwalt, Sr. Endowed Travel Fund,** University of South Carolina

2014 **Travel Grant,** Bangladesh-Sweden Trust Fund

2008–2009 **Dean's List Scholarship,** Bangladesh University of Engineering and Technology

2008–2009 **University Merit Scholarship,** Bangladesh University of Engineering and Technology

Certifications and Professional Development

2026–Present **Project Management Professional (PMP),** Project Management Institute

2021 **Postdoc Development Path,** ORNL Employee and Organizational Development

2016–Present **Engineer-in-Training (EIT),** South Carolina Board of Registration for Professional Engineers and Surveyors